KoopCast++: Neighbour-aware Koopman Pedestrian Trajectory Prediction

Design specification & v1 results

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Abstract. KoopCast++ predicts pedestrian futures with a *linear Koopman operator* acting on a lifted observable that already contains the (time-delayed) state, so *no decoder is needed*: the predicted positions appear directly in the observable and are read off by a fixed projection. The only learned component is a *social (neighbour) encoder*. The operator is obtained in closed form by *Extended DMD*, and the encoder is trained with a plain one-step Koopman *consistency* loss. The obstacle/environment branch of the earlier KoopyAdapt is dropped. Models are trained per held-out scene on the official ETH/UCY leave-one-out splits (Social-STGCNN distribution).

1 Setup

A scene has pedestrians $i = 1, \dots, N$ with positions $\mathbf{p}_i^t \in \mathbb{R}^2$ (world metres). Observation length T_o=8, prediction length T_p=12, Δt=0.4 s. At the prediction origin t_0 we observe $X_i = [\mathbf{p}_i^{t_0-T_o+1}, \dots, \mathbf{p}_i^{t_0}]$ and predict $[\mathbf{p}_i^{t_0+1}, \dots, \mathbf{p}_i^{t_0+T_p}]$.

2 Observable (lifted state)

The observable stacks a **time-delay history** of the state with a social embedding ($d = 2T_o + d_s = 16 + d_s$):

$$\boldsymbol{\psi}_i^t \;=\; \big[\; \underbrace{\mathbf{p}_i^t, \mathbf{p}_i^{t-1}, \dots, \mathbf{p}_i^{t-7}}_{16\ \text{(time-delay)}} ; \quad \underbrace{\mathbf{e}_i^t}_{d_s\ \text{social}} \;\big] \in \mathbb{R}^d.$$

Because the raw positions are coordinates of $\boldsymbol{\psi}$, advancing $\boldsymbol{\psi}$ moves the most-recent-position slot to the next step; the predicted position is recovered by a *fixed* read-out $C = [\, I_2 \; \mathbf{0} \,]$, i.e. $\hat{\mathbf{p}} = C\boldsymbol{\psi}$. **There is no learned decoder.**

Social encoding. Around ego i at time t, neighbours within radius $R = 2.0$ m are binned into a S=12-sector polar proximity histogram $\mathbf{g}_i^t \in \mathbb{R}^S$,

$$\mathbf{g}_i^t[k] = \max_{j:\angle(j)\in \text{sec } k,\, d_{ij}\le R} \Big(1 - \tfrac{d_{ij}}{R}\Big), \quad d_{ij} = \|\mathbf{p}_j^t - \mathbf{p}_i^t\|,$$

compressed by the learned MLP $\mathbf{e}_i^t = E_{\text{dyn}}(\mathbf{g}_i^t)$, $E_{\text{dyn}} : 12 \to 16 \to 16 \to d_s$ (default d_s=4).

3 Koopman propagation

A single linear operator $K \in \mathbb{R}^{d\times d}$ advances the observable, and the horizon is produced by iterating it:

$$\boldsymbol{\psi}_i^{t+1} = K\,\boldsymbol{\psi}_i^{t}, \qquad \hat{\mathbf{p}}_i^{\,t_0+\tau} = C\,K^{\tau}\boldsymbol{\psi}_i^{t_0}, \quad \tau = 1,\dots,T_p.$$

The social block is propagated by K as part of $\boldsymbol{\psi}$ (neighbours' futures are not recomputed during rollout).

4 Operator via Extended DMD

Stack the lifted training transitions column-wise, $\Psi = [\boldsymbol{\psi}^t]$, $\Psi' = [\boldsymbol{\psi}^{t+1}]$ over all consecutive pairs (all agents, all training scenes). EDMD gives K in closed form,

$$K \;=\; \Psi'\,\Psi^{\dagger} \;=\; \Psi'\Psi^{\top}\big(\Psi\Psi^{\top} + \lambda I\big)^{-1},$$

with a small ridge λ. The solve is differentiable in Ψ, hence in the encoder parameters.

5 Training objective (v1: plain consistency)

The only learnable parameters are the social encoder $\theta = \mathrm{params}(E_{\mathrm{dyn}})$, trained to minimise the one-step Koopman consistency residual:

$$\boxed{\mathcal{L}(\theta) = \frac{1}{|\mathcal{P}|}\sum_{(t)\in\mathcal{P}} \big\|\,\boldsymbol{\psi}^{\,t+1}(\theta) - K\,\boldsymbol{\psi}^{\,t}(\theta)\big\|_2^2, \quad K = K(\theta) \text{ via EDMD.}}$$

Since K is the EDMD minimiser for the current encoding, $\mathcal{L}$ is the minimal linear-fit residual; minimising it over θ pushes the lifted coordinates toward a Koopman-invariant (linearly evolvable) subspace (*cf.* LKIS, Takeishi et al. 2017). *Caveat:* a plain consistency loss admits a trivial collapse of the social block (a constant embedding is perfectly self-predictable); this v1 accepts the risk and we compare against alternatives (e.g. a state-block multi-step prediction loss) if needed.

Procedure. per minibatch (here: full batch): encode $\to \Psi, \Psi'$; solve EDMD for K; compute $\mathcal{L}$; backprop to θ. After training, refit one global K on all training pairs. One model per held-out scene.

6 Adaptive online operator (optional)

At inference an agent-specific operator can track the agent's recent dynamics by a streaming rank-one update over *observed* lifted states,

$$K_i \;\leftarrow\; K_i + \eta\,\frac{\big(\boldsymbol{\psi}^{\,s+1} - K_i\boldsymbol{\psi}^{\,s}\big)\,\boldsymbol{\psi}^{\,s\top}}{\|\boldsymbol{\psi}^{\,s}\|^2},$$

applied for each consecutive observed pair $(\boldsymbol{\psi}^s, \boldsymbol{\psi}^{s+1})$ before the forward rollout (η=0 recovers the static global K). The update must use *real* observations: applying it to the predicted rollout gives a zero residual (no-op), which is exactly why the legacy implementation's adaptation was inert. With the time-delay-8 observable and the standard 8-step protocol only one observable is available, so adaptation is effective only with a longer / streaming history; the mechanism is implemented (`eta`) and left off in the v1 benchmark.

7 Relation to predecessors

- **KoopCast**: Koopman + MDN decoder, k-nearest relative neighbours. (removed from the repo.)
- **KoopyAdapt**: adds an obstacle branch, sector social, online adaptation; uses encoders + a learned decoder.
- **KoopCast++ (this model)**: *no decoder* (state in the observable), *no obstacle branch*, social-only learned dictionary, EDMD operator, plain consistency loss, $R = 2\,\mathrm{m}$.

8 v1 results (ETH/UCY leave-one-out)

Same neighbour-aware target set per scene (`evaluate_scene`); ADE/FDE in metres, $8 \to 12$. Baseline is constant velocity (last observed velocity extrapolated).

	ConstantVelocity		KoopCast++ (static)	
scene	ADE	FDE	ADE	FDE
eth	1.076	2.282	**1.066**	**2.186**
hotel	**0.319**	**0.614**	0.499	1.087
univ	**0.604**	**1.339**	0.724	1.523
zara1	**0.427**	**0.952**	0.436	0.971
zara2	**0.322**	**0.724**	0.354	0.788
AVG	**0.550**	**1.182**	0.616	1.311

Reading. The v1 model (plain consistency loss) is on par with constant velocity on ETH (slightly better) and ZARA1, but worse on HOTEL/UNIV/ZARA2, so on average it does not yet beat the baseline. The model is well-behaved (operator spectral radius ≈ 1, no rollout blow-up), unlike the old KoopCast inference path. This is the expected v1 checkpoint; next iterations vary the training loss (state-block multi-step), social representation, and d_s, comparing back to this table.

9 Defaults / open knobs

- social embedding width d_s (default 4); E_{dyn} MLP size.
- EDMD ridge λ (default 10^{-4}); minibatch vs. global fit.
- alternative losses (state-block multi-step prediction) for the next iteration.
- online K adaptation (η) for streaming / longer-context use.